MCMC Part 2

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Overview

Today, we cover:

- Gibb's sampling
- Hamiltonian MC (briefly)

Resources:

- Peng Advanced Stat Computing Chapter 7
- Book chapter on Hamiltonian MC

Single component Metropolis-Hastings

The standard Metropolis algorithm updates the entire parameter vector at once with a single step.

- Hence, a p-dimensional parameter vector must have a p-dimensional proposal distribution q.
- However, it is sometimes simpler to update individual components of the parameter vector one at a time, in an algorithm known as single component Metropolis-Hastings (SCMH).

Let $x^t = (x_1^t, x_2^t, \dots, x_p^t)$ be a p-dimensional vector of parameter values at iteration t.

Single component M-H

The SCMH algorithm updates each parameter in x one at a time. All that is needed to update the ith component of x^t is a proposal distribution $q(x_i^*|x_i^t, x_{-i}^t)$. SCMH updates the ith component via the following steps:

(1) Sample $x_i^* \sim q(x_i^*|x_i^t, x_{-i}^t)$ as a proposal for component i.
(2) Let

$$\alpha(x_i^*|x^t) = \min\left\{\frac{\pi(x_i^*|x_{-i}^t)q(x_i^t|x_i^*, x_{-i}^t)}{\pi(x_i^t|x_{-i}^t)q(x_i^*|x_i^t, x_{-i}^t)}, 1\right\}$$

(3) Accept x_i^* with probability $\alpha(x_i^*|x^t)$
(4) Repeat steps 1-3 p times until each component is updated

Gibb's sampling

- Special case of single component M-H where you always accept the candidate
- For sampling from multivariate distributions (2 or more parameters)
  - Relies on ability to calculate full conditionals

Useful when:

- Sampling from $\pi(\theta_1, \theta_2, \theta_3)$ is hard
- Sampling from $\pi(\theta_1|\theta_2, \theta_3)$, $\pi(\theta_2|\theta_1, \theta_3)$, $\pi(\theta_3|\theta_1, \theta_2)$ is easy

Gibb's sampling algorithm

Steps for each iteration:

1. Sample $\theta_1^{n+1} \sim \pi(\theta_1 | \theta_2^n, \theta_3^n)$
2. Sample $\theta_2^{n+1} \sim \pi(\theta_2 | \theta_1^{n+1}, \theta_3^n)$
3. Sample $\theta_3^{n+1} \sim \pi(\theta_3 | \theta_1^{n+1}, \theta_2^{n+1})$

Do this for the desired number of samples, discard burn-in, and voila! You now have a sample from the joint distribution $\pi(\theta_1, \theta_2, \theta_3)$.

Example: Bivariate normal distribution

Suppose we want to simulate $x = (x_1, x_2)$ from a bivariate Normal distribution with mean $\mu = (\mu_1, \mu_2)$ and covariance matrix

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho \\ \rho & \sigma_2^2 \end{pmatrix}$$

we can make use of Gibbs sampling to simulate a Markov chain that will converge to a bivariate Normal.

Bivariate normal distribution

Goal is to simulate from joint distribution $\pi(x_1, x_2|\mu, \Sigma)$.

Complete conditionals are:

- $\pi(x_1|x_2, \mu, \Sigma) = N(\mu_1 + \rho\frac{\sigma_1}{\sigma_2}(x_2 - \mu_2), \sigma_1^2(1-\rho))$
- $\pi(x_2|x_1, \mu, \Sigma) = N(\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x_1 - \mu_1), \sigma_2^2(1-\rho))$

So, we sample

1. $x_1^{t+1} \sim N(\mu_1 + \rho\frac{\sigma_1}{\sigma_2}(x_2^t - \mu_2), \sigma_1^2(1-\rho))$
2. $x_2^{t+1} \sim N(\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x_1^{t+1} - \mu_1), \sigma_2^2(1-\rho))$

Bivariate normal distribution

```
gibbs_bivariate = function(x0, n = 1000,
                           mu = c(1, 5), rho = 0.5,
                           sigma1 = 1, sigma2 = 2){
  x1 = x0[1]
  x2 = x0[2]
  mu1 = mu[1]
  mu2 = mu[2]

  x = matrix(NA, nrow = n, ncol = 2)
  for(i in 1:n){
    x[i,] = c(x1,x2)

    x1 = rnorm(1, mean = mu1 + rho * sigma1/sigma2 * (x2-mu2),
               sd = sigma1 * sqrt(1-rho))
    x2 = rnorm(1, mean = mu2 + rho * sigma2/sigma1 * (x1-mu1),
               sd = sigma2 * sqrt(1-rho))
  }
  colnames(x) = c("x1", "x2")
  return(x)
}
```

Bivariate normal distribution

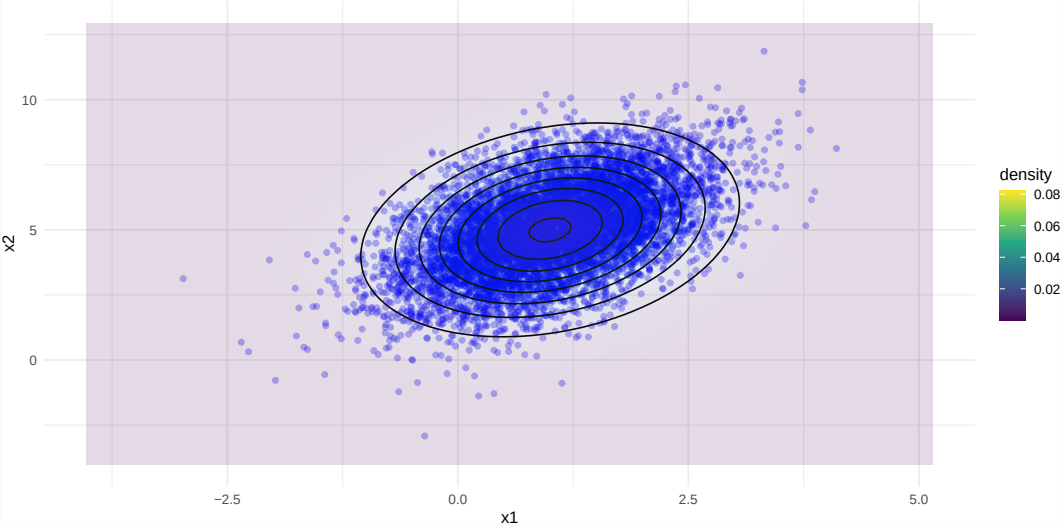
```r
set.seed(21433)
gibbs_x = gibbs_bivariate(c(0,0), n = 10000)

colMeans(gibbs_x[1:100,])
```

```
       x1        x2
0.7881165 4.6632079
```

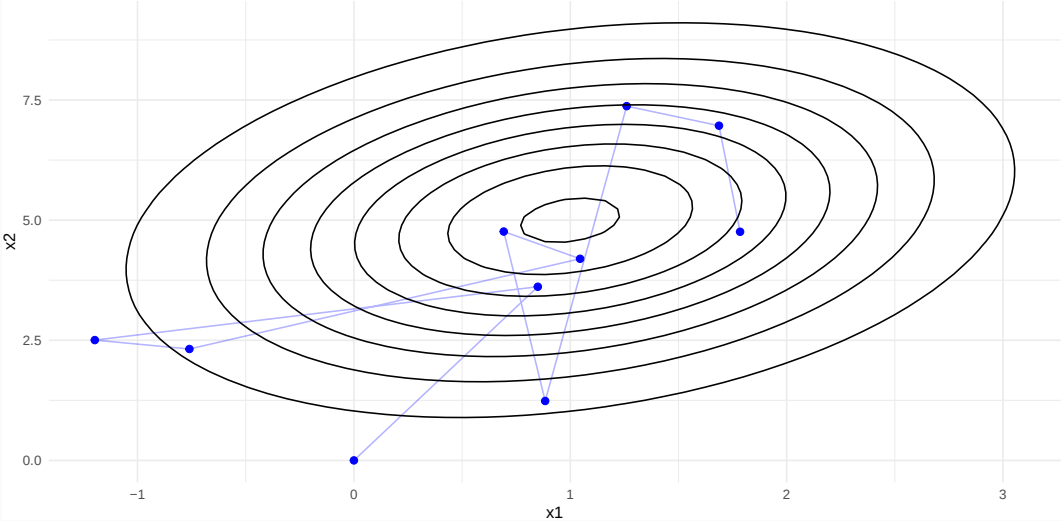
```r
colMeans(gibbs_x)
```

```
      x1       x2
1.012080 5.013594
```

Bivariate normal distribution



Bivariate normal distribution



Proof: Gibbs sampling as M-H

Gibbs is a special case of single component M-H that always accepts the proposal. Define the following:

- $\pi(\theta_1, \theta_2)$: joint distribution we want to sample from
- $q(\theta_1^* | \theta_1^t, \theta_2^t)$: proposal distribution for new value of θ_1
- Acceptance ratio:

$$\alpha(\theta_1^* | \theta^t) = \min \left\{ \frac{\pi(\theta_1^*, \theta_2^t) q(\theta_1^t | \theta_1^*, \theta_2^t)}{\pi(\theta_1^t, \theta_2^t) q(\theta_1^* | \theta_1^t, \theta_2^t)}, 1 \right\}$$

- Choose full conditional as proposal, i.e.:

Proof: Gibbs sampling as M-H

Bayesian linear regression

$$y = X\beta + \epsilon; \epsilon \sim N(0, \sigma_y^2 I_n)$$

- Need to make (prior) distributional assumptions about β, σ_y^2
  - $\beta \sim N(0, \sigma_b^2 I_p)$
  - $\sigma_y^2 \sim IG(A, B)$
- Joint posterior of interest is

$$\pi(\beta, \sigma_y^2 | y, x, A, B, \sigma_b^2)$$

- Need full conditionals
  - $\pi(\beta | \sigma_y^2, y, x, A, B, \sigma_b^2)$
  - $\pi(\sigma_y^2 | \beta, y, x, A, B, \sigma_b^2)$

Full conditionals

$$\beta \mid \sigma_y^2, y \sim \mathcal{N}(\beta_{\text{post}}, \Sigma_{\text{post}})$$

$$\Sigma_{\text{post}} = \left(\frac{1}{\sigma_y^2} X^\top X + \frac{1}{\sigma_\beta^2} I_p \right)^{-1}$$

$$\beta_{\text{post}} = \left(X^\top X + \frac{\sigma_y^2}{\sigma_\beta^2} I_p \right)^{-1} (X^\top y)$$

$$\sigma_y^2 \mid \beta, y \sim IG\left(A + n/2, B + \frac{1}{2} ||y - x\beta||^2 \right)$$

Drawbacks of Gibbs and Random walk MH

- Tend to get stuck in small regions of the posterior for a long time
  - Especially for large number of parameters
  - I.E. slow mixing for correlated parameters
  - Potentially low effective sample size
- Requires closed-form full conditionals
- Random walk Metropolis-Hastings can be slow because it slowly explores the state space

Variations on Gibb's sampling

- **Blocked Gibb's sampling**: update blocks (or groups) of parameters jointly rather than one at a time
  - Helps improve mixing when variables are highly correlated
  - We are already doing this for Bayesian linear regression
- **Collapsed Gibb's sampling**: integrate/marginalize out some variables analytically before sampling
  - Reduces dimensionality and can improve convergence
- **Metropolis-within-Gibbs**: Uses Metropolis-Hastings steps for full conditionals that are not easily sampled from.
  - Commonly used in Bayesian hierarchical models

Hamiltonian Monte Carlo

Though originating in physics, Hamiltonian dynamics can be applied to most problems with continuous state spaces by simply introducing fictitious "momentum" variables.

HMC alternates updates for momentum variables with Metropolis updates in which a new state is proposed by computing a trajectory according to Hamiltonian dynamics. A state proposed in this way can be distant from the current state but nevertheless have a high probability of acceptance.

Hamiltonian Monte Carlo

- The first step is to define a Hamiltonian function in terms of the probability distribution we wish to sample from.
- Parameters of interest are called “position” variables
- Also introduce auxiliary “momentum” variables

Hamiltonian dynamics

Hamiltonian dynamics has a physical interpretation that can provide useful intuitions. In two dimensions, we can visualize the dynamics as that of a frictionless puck that slides over a surface of varying height The state of this system consists of the position of the puck, given by a 2D vector q, and the momentum of the puck (its mass times its velocity), given by a 2D vector p. The potential energy, $U(q)$, of the puck is proportional to the height of the surface at its current position, and its kinetic energy, $K(p)$, is equal to $|p|^2/(2m)$, where m is the mass of the puck. On a level part of the surface, the puck moves at a constant velocity, equal to p/m. If it encounters a rising slope, the puck's momentum allows it to continue, with its kinetic energy decreasing and its potential energy increasing, until the kinetic energy (and hence p) is zero, at which point it will slide back down (with kinetic energy increasing and potential energy decreasing).

Hamiltonian dynamics

$$H(q,p) = U(q) + K(p)$$

- U(q): potential energy
- K(p): kinetic energy
  - Usually defined as $p^T M^{-1} p/2$, where M is a "mass matrix"

Hamiltonian dynamics

The partial derivatives of the Hamiltonian determine how q and p change over time t, according to Hamilton's equations:

$$\frac{dq_i}{dt} = \frac{\partial H}{\partial p_i}$$

$$\frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i}$$

Hamiltonian dynamics in statistics

- **position** corresponds to parameters of interest
- **potential energy U(q)** is the (joint) log probability density of interest
- **kinetic energy K(p)** is usually defined as $p^T M^{-1} p/2$, where M is a "mass matrix"
  - Corresponds to zero-mean Gaussian with covariance M

Can be used to construct MCMC updates.

Gibbs sampler for Bayesian regression

I started implementing this myself… for lab we will go the rest of the way there and compare it to the results of `stan_glm` from the Bayesian Data Analysis lecture.

- My partial Gibb's sampler is in the file `gibbs_blr.R` and the lab file is `lab_gibbs.Rmd`.
- Compare effective sample size for your Gibb's sampler and for Stan's HMC algorithm
- Compare diagnostic plots as well

Resources

- Book chapter on Hamiltonian MC
- Stan manual
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